



$$P \Leftarrow Q$$

$$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

Truth table :-

P	Q	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

of P and Q

One or the other are true, but both are not true.

$$(P \vee Q) \wedge \sim (P \wedge Q)$$

P	Q	$P \vee Q$	$P \wedge Q$	$\sim (P \wedge Q)$	$(P \vee Q) \wedge \sim (P \wedge Q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F